

CENG381 Stochastic Processes

Poisson Process — Practice 1

Q1 (30 points). Emails arrive at a server according to a Poisson process with rate $\lambda = 3$ emails per hour.

- a) Write the PMF of $N(t)$, the number of emails arriving in t hours. State also $E[N(t)]$ and $\text{Var}(N(t))$.
 - b) Compute $P(N(2) = 6)$ and $P(N(2) \leq 1)$. Leave your answers in exact form (powers of e) and also give decimal approximations.
 - c) Using the counting-process connection, express the event {first email arrives after time t } in terms of $N(t)$. Find $P(\text{first email arrives after } t = 1/2 \text{ hour})$.
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Q2 (35 points). Customers arrive at a coffee shop at rate $\lambda = 10$ per hour (Poisson). Each customer independently orders a latte with probability $p = 0.4$. Separately, a rival coffee truck parked nearby attracts customers at rate $\mu = 6$ per hour (independent Poisson).

- a) The latte-ordering process is a thinned Poisson process. State its rate. Find $P(\text{exactly 3 lattes ordered in 1 hour})$.
 - b) A health inspector monitors the total traffic — coffee shop plus truck combined. State the distribution and rate of the combined arrival process. Find $P(\text{at least 2 arrivals in 15 minutes})$.
 - c) It is now 9 AM and $N_{\text{shop}}(1) = 4$ (4 customers arrived in the first hour). Using the independent-increments property, find $E[N_{\text{shop}}(3) | N_{\text{shop}}(1) = 4]$.
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Q3 (35 points). This question connects the Poisson process to its differential-equation characterisation.

- a) From the infinitesimal description — $P(\text{one arrival in } (t, t+\Delta)) = \lambda\Delta + o(\Delta)$ and $P(\text{two or more arrivals in } (t, t+\Delta)) = o(\Delta)$ — derive the Kolmogorov forward equation:

$$P_j'(t) = -\lambda P_j(t) + \lambda P_{j-1}(t)$$

where $P_j(t) = P(N(t) = j)$.

- b) Verify that $P_j(t) = (\lambda t)^j \cdot e^{-\lambda t} / j!$ satisfies the forward equation. (Differentiate with respect to t and substitute.)

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- c) A Poisson process with rate $\lambda = 2$ models bus arrivals at a stop. You arrive at the stop at time 0. Find $P(\text{you wait more than 45 minutes for the first bus})$. Also compute the expected wait. Express your answer in exact form and give a decimal approximation.